P510/1
PHYSICS
PAPER 1
July/August 2026
3 Hours

ASSHU KIGEZI REGIONAL MOCK EXAMINATIONS
Uganda Advanced Certificate of Education

PHYSICS PAPER 1

Time: 3Hours

INSTRUCTIONS TO CANDIDATES

- *Answer 4 items only from both sections A and B.*
- *Section A has two parts with two items. Part I (motion and force) and part II (particles) which are all compulsory.*
- *Section B has also two parts with four items. Part I (energy) has two items and part II (charges and fields) has also two items. A candidate is supposed to answer one item from each of the parts I and II.*
- *Any additional item(s) attempted will not be marked.*
- *Begin each item on a fresh page.*
- *Silent non-programmable calculators may be used.*

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SECTION A

PART I: MOTION AND FORCE

ITEM 1:

In rural areas, communities often face water scarcity during dry seasons. Maria and her family depend on water from a raised local reservoir tank to irrigate their crops. She uses a manual pump that requires pulling a rope to activate a water-lifting mechanism. The pump's operation depends on a specific circular pulling motion and she wants to optimize her pumping speed without breaking the connecting components.

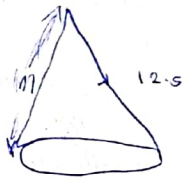
Maria ties a stone of mass 0.5kg to a strong nylon string of length 1m. She hangs the string from a fixed point 2m above the ground. She whirls the stone in a horizontal circle with an increasing spinning rate. She noted that the string will break when the tension reaches its limit and which happens when the angle with the vertical is at its maximum.

The pump's mechanism can be damaged if the string breaks during operation. Maria needs to determine the exact moment of failure to calibrate her manual pumping speed. She needs to calculate the physical limits of the system and find out where the stone lands if the string breaks, so she can safely operate the mechanism.

A learner is needed to describe the forces acting on the stone while it is whirled in a horizontal circle and determine the horizontal distance the stone travels from the point directly beneath the suspension if the string breaks at the instant of maximum tension.

Support materials;

- Acceleration due to gravity g = 9.81ms^{-2}
- Mass of the stone = 0.5kg
- Length of the string = 1m
- Tension in the string = 12.5N



Task

As a physics student;

(a) Analyze the data and provide the necessary assistance.

(b) Help Maria explain how she can apply the above principles to safely improve the water-pumping mechanism without breaking the nylon string.

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PART 11:

PARTICLES

ITEM 2:

You are part of a physics research team at Makerere University tasked with designing a safe and functional nuclear power plant to generate electricity using Uranium. Your supervisor challenges you with specific technical tasks that the team must solve.

The plant utilizes a metal door operated by a photoelectric sensor. The metal door has a work function of 2.2ev. As a learner determine if an incident radiation of wavelength 400nm can cause electron emission to trigger the door's operation. Inside the nuclear reactor, uranium -235 undergoes nuclear reaction, splitting into Barium -141 and krypton -92 along with two free neutrons. The supervisor requires that 1g of uranium must provide atleast 7.5×10^{10}J of energy to make the project economically viable.

A learner is tasked to explain how the electricity is generated inside a nuclear reactor and how dangerous nuclear wastes produced by the plant should be disposed of safely.

Support materials

- Avogadro's number NA = 6.02×10^{23} mol^{-1}
- Speed of light c = 3.0×10^{8} ms^{-1}
- Mass of Uranium $\left({}^{235}_{92}U\right) = 235.044\ u$
- Mass of Barium $\left({}^{141}_{56}Ba\right) = 140.914\ u$
- Mass of Krypton $\left({}^{92}_{36}Kr\right) = 91.926\ u$
- Mass of neutron $\left({}^{1}_{0}\cap\right) = 1.009\ u$
- 1 u = 931 Mev
- 1 ev = 1.6×10^{-19} J
- Plank's constant h = 6.63×10^{-34} JS

$\frac{m}{M} \times NA \cdot = \Delta m \cdot$

Task

As a physics student;

(a) Analyze the data and provide the necessary assistance.
(b) Help the supervisor to address his concerns.

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SECTION B

PART I: ENERGY

(Respond to one item from this part)

ITEM 3:

In our daily lives, we experience sound in many ways such as listening to music, hearing car horns or communicating. Sound travel through the air in waves that carry energy. When sound reaches our ears, it pushes against the eardrum making it vibrate back and forth.

Captain Amaru turns on a loud warning siren on his ship. The siren produces loud sound waves that travel through the air to the shore. The energy from this sound wave reaches Maya and causes her eardrum to vibrate.

The distance from the ship to the shore is 500m and the speed of sound is $335ms^{-1}$. The equation for the siren's wave displacement is

$y = r\sin 2\pi(100t - \frac{x}{y})$.

Earlier Maya was told that if the energy from sound wave reaching her eardrum exceeds $2.0 \times 10^{-16}J$, the sound is likely to damage her eardrum.

Maya wants to understand how the siren's sound reaches her ear, how much energy the sound wave carries and how long it takes for the sound to travel the distance between the ship and the shore.

Support Material

- Mass of Eardrum = $1.00 \times 10^{-5}kg$
- Amplitude = $1.00 \times 10^{-8}m$

Task

As a physics student;

(a) Analyze the data and provide the necessary help.

(b) Help to explain the device that uses some principles of vibration and echoes and suggest a tool a person can use to hear or measure sound more clearly if Maya was much further away and could only hear a very faint echo of the ship's siren.

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ITEM 4:

In mountainous areas of Western Uganda, farmers often use local materials such as clay bricks, blackened metals and glass lenses to construct simple un-powered solar dryers for preserving crops like beans and maize.

To ensure the crops dry well without burning, a local technician wants to use a simple optical pyrometer to monitor the temperature of a balanced clay plate absorbing sunlight.

Mr Kato a Physics teacher, sets up a simple laboratory demonstration to show how the sun's extreme temperature determines the heat felt on earth.

Learners assist in taking temperature readings, observe wave ripples in a nearby water basin and measure vibrations of a swinging pendulum.

Learners want to understand how much solar heat reaches a local drying rack and how to safely measure the temperature of the sun, determine the amount of solar heat (intensity) that reaches the earth's surface if the sun is 1.5×10^{11}m away.

If Pluto's average distance from the sun is about 40 times that of the earth's. Learners need to know the heat intensity received at Pluto's compared to earth and find out whether a crop-drying rack could function properly on Pluto.

Support Materials

- Temperature of the sun = 6000k
- Stefan – Boltz momn constant $\sigma = 5.67 \times 10^{-8}$ $Wm^{-2}k^{-4}$
- The sun's radius $r = 7.0 \times 10^{8}$m
- Orbital radius of the earth = 1.5×10^{11}m

Task

As a physics student;

(a) Analyze the data clearly and give the necessary assistance.
(b) Help Mr. Kato to explain how different wavelengths of light determine the color and perceived temperature of the sun.

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Part II: Charges and Fields

(Respond to one item from this part)

ITEM 5:

Sarah lives in a village and wants to set up a small poultry incubator. She has a car battery which provides direct current (DC) but her incubator needs alternating current (AC) to safely run the heating system. She hires Mark, a local electrician to design a circuit. Mark connects the battery to an inverter which changes the direct current (DC) to alternating current (AC) to power the incubator's heating load.

Mark fastened the battery up to the incubator. The battery has an unknown electromotive force (e.m.f) and an internal resistance. It is connected in series with the load (the incubator). A very high resistance voltmeter connected across the load reads 3.2v. Mark notices that 0.032w of power is lost as heat inside the battery itself and the circuit's overall efficiency is 80%.

Sarah wants to know if the battery system is working efficiently. She needs to understand how to maximize the power going to her incubator while making sure no parts overheat and that the alternating current (AC) system is stable.

Mark advised Sarah that for the system to work more effectively, the current flowing through the circuit should not be less than 0.038A

A learner is tasked to explain why does the voltmeter read 3.2v instead of the battery's full electromotive force and determine the electromotive force (e.m.f) of the battery.

Support Material

- Power = 1V
- Power in internal resistance = I^2r
- Efficiency = $\frac{Useful\ power}{total\ power}$ X 100% or
- Efficiency = $\frac{Load\ Voltage}{total\ e.m.f}$ X 100%

Task.

As a physics student;

(a) Analyze the data and provide the necessary assistance

(b) Help Sarah to explain how the device works safely and determine the step-up ratio of the device if the heating load of her incubator draws 0.04A of current and the device in the inverter steps up voltage from her battery to a 240V AC.

ITEM 6:

Musa lives in a village where fetching water for a community garden is hard work. To make daily life easier, he builds a small water-lifting system. This system uses an electric motor that runs on a charged capacitor instead of a battery.

Musa fully charges the capacitor using an external power source. He then disconnects the power source and connects the capacitor to the small electric motor so that it can lift a water bucket.

To get water from the well, the motor needs to lift a mass of 150g. However, Musa notices that the system is not perfectly efficient and only 20% of the stored electrical energy gets used to lift the water load.

A learner needs to explain how does a capacitor store electrical energy and what is its role in the system when Musa connects it to the motor.

If Musa upgrades his setup to include an ac circuit running on a coil of a device to step-up the voltage, explain how would that affect the capacitor.

The village engineer told Musa that for the system to work more effectively, the actual useful mechanical work done to lift the load, should not be less than 0.255J.

Using the useful mechanical energy determined, a learner is tasked to find out how high the motor can lift the 150g load in one complete discharge of the capacitor.

Support Materials

- A Voltmeter
- Connecting Wires
- A 500μf Capacitor
- A 24v Power Source
- An Electric Motor
- Acceleration due to gravity $g = 9.81 ms^{-2}$

TASK

As a physics student;

(a) Analyze the data and provide the necessary assistance
(b) Help Musa to explain how the concept of capacitors discharging to run motors is similar to devices used in our homes such as camera flashes.

——— *End* ———

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